

EV Charging Station Management System

**UCS310 - Database Management Systems
B. Tech (2nd Year) - COE 2025-2026**



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

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INTRODUCTION

The EV Charging Station Management System is designed to manage charging stations, chargers, bookings, charging sessions, and payments using a centralized database. It helps users check charger availability, reserve slots, and track charging usage efficiently.

The motivation for this project comes from increasing EV adoption and the operational challenges at charging stations. Manual tracking of bookings and billing leads to delays, resource underutilization, and service conflicts during peak demand periods.

A Database Management System (DBMS) is preferred over file-based tracking because it provides structured storage, faster data retrieval, secure multi-user access, and reliable transaction handling for booking and billing workflows.

PROBLEM STATEMENT

Many charging stations face slot conflicts, uncertain availability, and inconsistent billing due to fragmented or manual record keeping. Users often arrive without real-time visibility of available chargers, causing queues and poor service experience.

Without integrated data management, operators struggle to maintain accurate session logs, tariff calculations, payment status, and maintenance records. This creates reporting gaps and reduces operational transparency.

The proposed DBMS solution stores station, charger, booking, session, and payment data in relational tables with constraints and transaction control to ensure consistency, reduce conflicts, and improve efficiency.

OBJECTIVES OF THE PROJECT

1. To design a proper database for the EV Charging Station Management System using E-R data modelling.
2. To convert the ER model into relational tables with suitable primary keys and foreign keys.
3. To apply normalization techniques up to Third Normal Form (3NF) or BCNF to reduce redundancy and improve efficiency.
4. To implement the database using SQL commands for creating, inserting, updating, and retrieving data.
5. To use PL/SQL constructs such as procedures, functions, triggers, and cursors to automate booking, charging session, and billing operations.
6. To ensure data consistency, accuracy, and security by using constraints and transaction management.
7. To enforce relationship constraints such as one-to-one and one-to-many mappings using keys and unique constraints.

SCOPE OF THE PROJECT

Functional Scope:

- Maintain charging station and charger master data.
- Allow users to search and book charging slots.
- Capture session lifecycle from start to completion.
- Apply tariff rules and generate billing records.
- Track payment status and session history.
- Generate utilization, revenue, and maintenance reports.

Out of Scope:

- Real-time IoT charger hardware control.
- Battery condition analysis.
- GPS-based vehicle tracking.

Types of Users Involved:

- Admin/Operator – Manages stations, chargers, tariff plans, and maintenance logs.
- Registered User – Books slots, initiates sessions, views billing history, and makes payments.
- Guest/Viewer – Browses station and charger availability without booking.

Modules Covered (Backend Focus):

- Station & Charger Management – DDL/DML for station and charger master tables.
- Booking Module – Slot reservation with trigger-based conflict prevention.
- Session Management – Session lifecycle tracking via stored procedures.
- Payment & Billing – Tariff-based amount calculation using PL/SQL functions.
- Reporting – Revenue and utilization queries using cursors and views.
- Maintenance Logging – Charger issue tracking and resolution records.

PROPOSED SYSTEM DESCRIPTION

The proposed system manages all EV charging operations using a centralized database. Users can browse available chargers, book time slots, start sessions, and make payments. When a session begins, the system records start time, charger ID, and user ID. After the session ends, the system calculates energy consumed and applies tariff rules to compute payable amount.

Triggers prevent double booking, procedures update charge status and generate billing entries, and transactions ensure safe execution of booking and payment workflows. Operators and administrators can monitor live charger availability, session progress, tariff plans, and maintenance records.

The system also manages tariff plans for dynamic pricing and maintains maintenance logs for chargers to track faults and servicing history, ensuring operational reliability.

DATABASE DESIGN

The database of the EV Charging Station Management System is designed to store and manage all charging operations in a structured and normalized format. It supports fast retrieval, integrity, and scalable multi-station management.

All one-to-one relationships are enforced using UNIQUE constraints on foreign keys.

ENTITY-RELATIONSHIP (ER) DIAGRAM

Entities Identified:

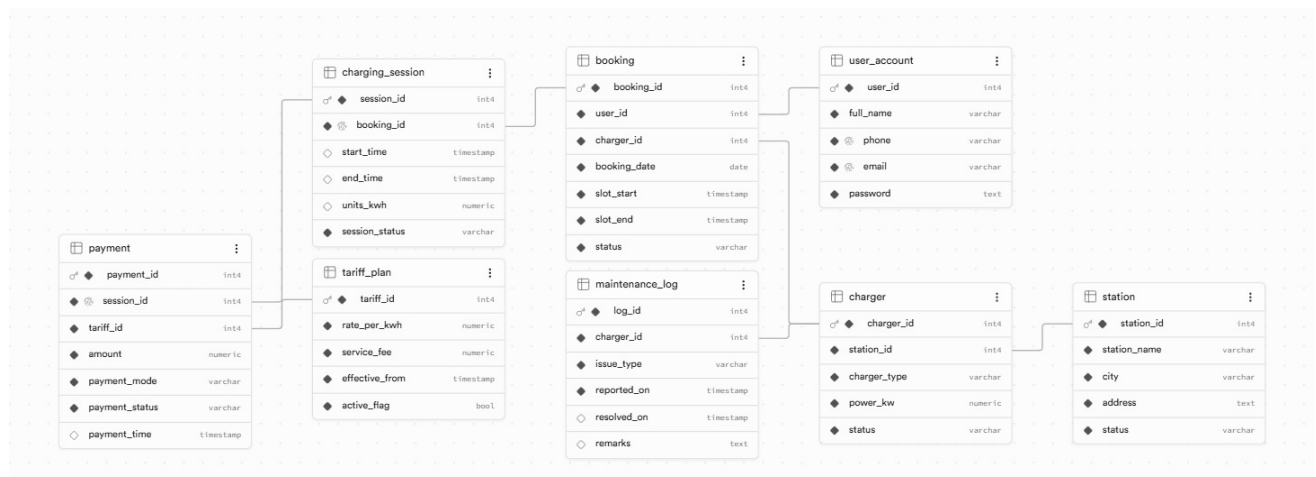
Station, Charger, User_Account, Booking, Charging_Session, Tariff_Plan, Payment, Maintenance_Log

Key Attributes:

- Station: station_id (PK), station_name, city, address, status
- Charger: charger_id (PK), station_id (FK), charger_type, power_kw, status
- User_Account: user_id (PK), full_name, phone (UNIQUE), email (UNIQUE), password
- Booking: booking_id (PK), user_id (FK), charger_id (FK), slot_start, slot_end, status
- Charging_Session: session_id (PK), booking_id (FK UNIQUE), start_time, end_time, units_kwh
- Payment: payment_id (PK), session_id (FK UNIQUE), tariff_id (FK), amount, payment_mode, payment_status

Relationships & Cardinality:

- Station – Charger: One-to-Many (one station has many chargers)
 - User_Account – Booking: One-to-Many (one user can have many bookings)
 - Charger – Booking: One-to-Many (one charger can be booked multiple times)
 - Booking – Charging_Session: One-to-One (each booking leads to one session)
 - Charging_Session – Payment: One-to-One (each session generates one payment)
 - Charger – Maintenance_Log: One-to-Many (one charger may have multiple maintenance entries)
- (ER diagram shown below)



Relational Schema

- Station(station_id PK, station_name, city, address, status)
- Charger(charger_id PK, station_id FK, charger_type, power_kw, status)
- User_Account(user_id PK, full_name, phone UNIQUE, email UNIQUE, password)
- Booking(booking_id PK, user_id FK, charger_id FK, booking_date, slot_start, slot_end, status)
- Charging_Session(session_id PK, booking_id FK UNIQUE, start_time, end_time, units_kwh, session_status)
- Tariff_Plan(tariff_id PK, rate_per_kwh, service_fee, effective_from, active_flag)
- Payment(payment_id PK, session_id FK UNIQUE, tariff_id FK, amount, payment_mode, payment_status, payment_time)
- Maintenance_Log(log_id PK, charger_id FK, issue_type, reported_on, resolved_on, remarks)

NORMALIZATION

Normalization organizes data to reduce redundancy and improve consistency. In this project, it ensures an efficient and well-structured EV charging operations database.

Functional Dependencies

Some important functional dependencies in the system are:

- station_id -> station_name, city, address, status
- charger_id -> station_id, charger_type, power_kw, status
- user_id -> full_name, phone, email, password
- booking_id -> user_id, charger_id, booking_date, slot_start, slot_end, status
- session_id -> booking_id, start_time, end_time, units_kwh, session_status
- tariff_id -> rate_per_kwh, service_fee, effective_from, active_flag
- payment_id -> session_id, tariff_id, amount, payment_mode, payment_status, payment_time
- log_id -> charger_id, issue_type, reported_on, resolved_on, remarks
- phone -> user_id (assuming phone is unique)

email -> user_id (assuming email is unique)

These dependencies help in identifying how attributes depend on primary keys.

Normalization Process

- **1NF: Atomic values, no repeating groups.**
- **2NF: No partial dependency; non-key attributes fully depend on primary key.**
- **3NF: No transitive dependency; non-key attributes depend only on the primary key.**
- **BCNF: Every determinant is a candidate key.**

Final Normalized Structure

The final normalized database structure (after applying 1NF, 2NF, 3NF, and BCNF where applicable) is as follows:

Station

(station_id, station_name, city, address, status)

Charger

(charger_id, station_id, charger_type, power_kw, status)

User_Account

(user_id, full_name, phone, email, password)

Booking

(booking_id, user_id, charger_id, booking_date, slot_start, slot_end, status)

Charging_Session

(session_id, booking_id UNIQUE, start_time, end_time, units_kwh, session_status)

Tariff_Plan

(tariff_id, rate_per_kwh, service_fee, effective_from, active_flag)

Payment

(payment_id, session_id UNIQUE, tariff_id, amount, payment_mode, payment_status, payment_time)

Maintenance_Log

(log_id, charger_id, issue_type, reported_on, resolved_on, remarks)

The normalized design removes redundancy, enhances performance, and ensures accurate, consistent, and reliable operations across all multi-station EV charging workflows.

DATABASE IMPLEMENTATION

The system uses SQL and PL/SQL components to manage and automate EV charging operations.

SQL Implementation

DDL:

- CREATE TABLE for all entities.
- ALTER for modifying constraints.
- DROP for schema reset.

DML:

- INSERT new bookings and sessions.
- UPDATE charger status.
- DELETE cancelled reservations.

SELECT Queries:

- Joins: user–booking–session linkage across tables.
- Subqueries: find users with unpaid sessions, chargers with no bookings.
- Aggregate functions: SUM(amount), COUNT(booking_id), AVG(units_kwh).
- GROUP BY / HAVING: revenue grouped by station, sessions grouped by charger type.
- Views: ChargerAvailabilityView, UserBillingSummaryView, StationRevenueView.

PL/SQL Components:

- Stored Procedures: CreateBooking, StartChargingSession, CloseSession.
- Functions: CalculatePayableAmount, CheckChargerAvailability.
- Triggers: PreventDoubleBooking (BEFORE INSERT on Booking) and enforce one active session per booking.
- Cursors: Monthly revenue report iterating over session-payment records.
- Exception Handling: WHEN NO_DATA_FOUND, WHEN OTHERS for invalid operations.

TRANSACTION MANAGEMENT & CONCURRENCY

Transactions ensure safe multi-step updates.

COMMIT makes changes permanent.

ROLLBACK restores previous state on failure.

SAVEPOINT offers partial rollback.

ACID Properties:

Atomicity, Consistency, Isolation, Durability.

Concurrency:

Locking and isolation levels prevent lost updates and dirty reads.

TOOLS & TECHNOLOGIES USED

- **DBMS: Oracle / MySQL / PostgreSQL**
- **Query Language: SQL**
- **Programming: PL/SQL**
- **Tool: Oracle SQL Developer / MySQL Workbench**
- **OS: Windows / Linux**

The DBMS manages centralized charging data, while SQL and PL/SQL support implementation, automation, and reporting.

EXPECTED OUTCOMES

- **Normalized and efficient EV charging database design**
- **Accurate station availability, booking, and billing data retrieval**
- **Automated booking-to-session-to-payment workflow**
- **Reduced manual errors and charging slot conflicts**
- **Improved consistency, transparency, and data integrity.**